■ Blue Team Lab Log & Practical Portfolio

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Focus Area: Defensive Security / Blue Team Operations

1. Lab Environment Architecture

To perform safe analysis without endangering the host operating system or local network, an isolated malware analysis platform was constructed:

• **Hypervisor:** Oracle VM VirtualBox running on a Windows 11 host.

• **Guest Operating System:** REMnux (Noble Linux distribution built specifically for malware reverse engineering and security analysis).

• **Network Security Shield Configuration:** Network Interface Card (NIC) set exclusively to **Host-only Adapter** mode. This ensures complete network isolation, preventing any executing artifacts from accessing the physical local area network or the public internet.

2. Directory & System Commands Discovered

pwd (Print Working Directory)

Purpose: Displays the absolute path of the directory the analyst is currently working within.

Lab Output: /home/remnux

ls -la (List All - Long Format)

Purpose: Lists all files and subdirectories. The -l flag displays file details (permissions, owner, size, date). The -a flag uncovers hidden files starting with a dot (.), a critical technique for discovering hidden malware scripts.

Linux Permissions Matrix Structure

Observed string architecture during lab logging: drwxr-xr-x vs -rw-r--r--

• **First Character Indicator:** 'd' = Directory (Folder), '-' = Regular Data File (e.g., .bashrc)

• **Security Access Blocks:** 'r' = Read, 'w' = Write (edit), 'x' = Execute (run program)

grep -i "keyword" [filename]

Purpose: Acts as a high-speed keyword scanner inside text files. The -i flag ignores case-sensitivity.

Defensive Use Case: Used to quickly isolate malware indicators (IP addresses, rogue domains) from massive logs without reading line-by-line.

top (Table of Processes)

Purpose: Real-time system diagnostics interface (Linux Task Manager). Monitors CPU/Memory allocation. High CPU usage combined with low Idle space (id) triggers an investigation for unauthorized processes (like cryptojacking or active ransomware).

ss -tulpn (Socket Statistics)

Purpose: Displays an instant inventory of active network connections and listening ports. Filters specifically for TCP, UDP, Listening ports, displaying the Program name and raw Numeric port numbers.

3. Network Traffic Sniffing (Wireshark Laboratory)

Execution Command: wireshark &

Purpose: Launches the Wireshark GUI packet analyzer tool. The & symbol forces the program to execute in the background, keeping the parent terminal open.

Protocol	Description	OSI Layer
DHCP	Dynamic Host Configuration Protocol. Automatically handles IP address assignments to devices.	Layer 7 (Application)
ARP	Address Resolution Protocol. Maps IP addresses to physical MAC hardware addresses.	Layer 2 (Data Link)

Active Lab Analysis Exercise:

During packet sniffing on interface **enp0s3**, a traffic capture was frozen using the Stop (Red Square) control:

1. **Hardware Identification:** Successfully isolated the physical interface identity of the testing machine through an ARP broadcast: PCSSystemtec_c6:ee:...

2. **Traffic Filtering Operations:** Applied the display filter 'arp' into the Wireshark filter console. This instantly purged background DHCP broadcast noise, isolating specific Layer 2 validation logs.